Distance Travelled as a Signal of Talent

Short research memo: literature, empirical agenda, and firm heterogeneity

September 2026

Research question. Do candidates who have progressed unusually far relative to their starting opportunities carry information about latent talent that conventional credentials miss? The project treats "distance travelled" as a testable predictive signal, not a normative bonus: the goal is to construct an opportunity-adjusted measure from information available at application, validate it against later performance, and test whether employers use it efficiently.

1. Literature review: four strands

1.1 Positive selection and the meaning of barriers

The core economic intuition comes from selection. Mulligan and Rubinstein (2008) and Olivetti and Petrongolo (2008) show that the composition of observed workers depends on participation and nonemployment selection. Ashraf et al. (2025) bring this logic into the firm: where female labor-force participation is lower, observed female workers are more positively selected and outperform relative to men inside the same multinational. For distance travelled, the implication is that crossing a higher barrier can contain information about the latent type of the person who crosses it. The empirical challenge is to translate that group-level logic into an individual signal available before hiring.

1.2 Context changes the interpretation of signals

Behaghel, Crépon, and Le Barbanchon (2015) is the central paper for the information mechanism. In their anonymous-resume experiment, suppressing identifying information reduced minority candidates' interview chances in the selected firms because context had helped recruiters interpret other CV signals, including employment interruptions. Agan and Starr (2018) similarly show that removing individual information can induce employers to substitute group-level beliefs. Bartoš et al. (2016) add an attention mechanism: employers may process otherwise useful information asymmetrically. Together, these papers motivate a standardized context-adjusted signal rather than either raw background disclosure or complete blinding.

1.3 Employers learn latent ability slowly

Employer-learning models make a better ex ante signal valuable. Farber and Gibbons (1996), Altonji and Pierret (2001), Lange (2007), and Arcidiacono, Bayer, and Hizmo (2010) show that firms begin with imperfect information and update as performance becomes observable. Pallais (2014) demonstrates an important consequence: inexperienced workers can be inefficiently excluded because employers do not capture the full value of information generated by giving someone a chance. A distance-travelled measure attempts to move some of that learning to the hiring stage.

1.4 Data-driven screening and exploration

Personnel economics also shows that structured decision aids can outperform discretionary screening. Autor and Scarborough (2008) find that test-based screening improved retention without reducing minority hiring, while Hoffman, Kahn, and Li (2018) find that managers who override test recommendations tend to make worse hires. Li, Raymond, and Bergman (2026) model hiring as exploration and show that an exploration-aware algorithm can improve hiring yield while expanding the interview pool. Kessler, Low, and Sullivan (2019) provide a useful experimental

method for eliciting employer preferences over resume attributes. These studies suggest that distance travelled could enter either as a predictive feature or as a structured "second-look" signal for candidates whose paths are underrepresented in historical hiring data.

2. Literature map

Paper / practice	Core result	Connection to distance travelled	What remains missing
Ashraf et al. (2025)	Higher female performance where women are underrepresented; consistent with positive selection under barriers.	Harder entry can imply stronger latent types among those observed.	Individual-level ex ante measure from application data.
Behaghel et al. (2015)	Anonymous resumes reduced minority interview rates in selected firms; context changed how CV signals were interpreted.	The same observed signal can mean something different depending on the circumstances that generated it.	No progress measure and no validation against future productivity.
Smith & Naylor (2005)	Conditional on prior attainment, independent-school students were less likely to earn a good university degree than state-school students.	Endpoint credentials can be more informative when interpreted relative to prior environment.	Education setting rather than employer hiring.
Bastedo et al. (2022)	Context dashboards changed admissions recommendations for low-SES applicants.	Evaluators respond to structured opportunity context.	Admissions setting; no worker-productivity outcome.
Contextual recruitment / Rare CRS	Existing systems use disadvantage flags and academic outperformance relative to school context.	Very close implementation analogue.	Limited peer-reviewed causal validation of later job performance.
Pallais (2014)	Hiring and richer evaluations reveal information and improve later employment for inexperienced workers.	Motivates identifying overlooked talent before the market has learned ability.	Information is generated after work begins.
Li et al. (2026)	Exploration-aware screening can improve hiring quality and representation relative to human screening.	Framework for candidates with uncertain but potentially high upside.	Does not explicitly model starting-point-adjusted achievement.
Kessler et al. (2019)	Incentivized resume ratings recover employer preferences over candidate attributes.	Natural design to test whether employers underweight contextual achievement.	Method rather than a direct distance-travelled result.

3. Empirical agenda

The empirical agenda should separate four questions: whether the signal predicts performance, whether employers underweight it, whether a structured intervention improves selection, and whether firms differ systematically in their willingness or ability to hire high-distance-travelled candidates.

3.1 Does distance travelled predict productivity?

Start with retrospective employer data linking information observable at application to later outcomes such as structured interview scores, training performance, retention, promotion speed, manager evaluations, sales, or role-specific productivity. The key test is whether an opportunity-adjusted distance measure, D_i, adds predictive content beyond conventional credentials and other application controls. The economically relevant object is the out-of-sample gain from adding D_i: lower prediction error, better calibration, improved ranking quality, or better performance among marginal candidates selected.

Selective labels are the central econometric problem. Later productivity is observed only for people the firm already hired. Possible solutions are to use pre-hire work samples or standardized assessments available for a wider applicant pool; exploit quasi-random recruiter assignment or recruiter leniency; or deliberately interview or hire a randomized subset of high-distance-travelled marginal candidates.

3.2 Do employers underweight the signal?

A clean experiment would hold the conventional endpoint fixed while randomizing credible information about the starting point or opportunity set. Matched profiles could have the same GPA and degree but differ in school-level performance context, verified local opportunities, or merit-plus-need scholarship status. The main estimands are the causal effect of contextual information on interview interest; whether recruiters place less weight on it than its predictive value for later performance; and whether a standardized context score performs differently from raw background disclosure. This design directly tests the idea from Behaghel et al. (2015) that the same CV feature may carry different information in different contexts.

3.3 Field RCT of a "second-look" tool

A practical field experiment could randomize vacancies or recruiters across three arms: (1) business as usual; (2) a standardized context panel shown next to conventional credentials; and (3) a second-look recommendation that flags high-D candidates near the interview cutoff for additional review or a work sample. Outcomes should include interview rates, offers, acceptance, and ex post performance. The informative test is whether the intervention improves the quality of selected marginal candidates, or maintains performance while expanding the set of candidates who receive serious consideration.

3.4 Firm heterogeneity: which firms hire more "distant" candidates, and why?

A central extension is to treat firms themselves as heterogeneous screening technologies. Some firms may systematically identify and hire candidates whose conventional credentials look weaker but whose trajectories imply unusually large distance travelled. The first descriptive object would be a firm-level propensity to hire high-D candidates, estimated conditional on occupation, location, experience level, vacancy characteristics, and - where possible - the composition of the applicant pool. The goal is then to ask what distinguishes firms with persistently high versus low propensities.

- **Screening and HR sophistication:** Firms with structured assessments, richer applicant data, dedicated recruiting teams, or more standardized evaluation may be better able to extract signal from nontraditional paths.

- **Experience and organizational learning:** Firms that have previously hired high-D candidates may learn that conventional credentials underpredict their performance and update future screening rules.
- **Tolerance for uncertainty and exploration:** Some firms may be more willing to interview candidates with noisier conventional signals or deliberately explore outside familiar talent pools.
- **Jobs and production technology:** Distance travelled may be more useful in roles where adaptability, learning, and growth matter more than immediately verifiable task-specific credentials.
- **Recruiter incentives and decision rights:** Hiring managers facing strong short-run downside from a bad hire may rely more heavily on conventional prestige signals, while firms that can absorb experimentation or use probationary/work-sample stages may screen differently.
- **Labor-market conditions:** Tight labor markets or difficulty filling vacancies may induce firms to search more broadly and learn about candidates outside standard pipelines.

The strongest version of this analysis would connect hiring behavior to outcomes. Do high-D-hiring firms obtain higher productivity, faster promotion, lower turnover, or better performance specifically among nontraditional candidates? Do firms change their screening behavior after unexpectedly successful high-D hires? This turns firm heterogeneity from a descriptive fact into a mechanism: some organizations may be better at discovering latent talent, and the project can study which organizational practices generate that advantage.

4. What appears genuinely open

The existing literatures cover nearly every ingredient separately: barriers can generate positive selection; the informational content of CV features depends on context; firms learn ability after hiring; contextual admissions and recruitment already adjust achievement for opportunity; and structured screening or exploration can improve hiring decisions.

The clearest open contribution is at their intersection. A research paper could construct an opportunity-adjusted, trajectory-based signal from ordinary application data, validate its incremental predictive content for worker performance, and experimentally identify whether employers systematically underuse it. Adding firm heterogeneity sharpens the contribution further: rather than treating all employers as having the same screening rule, the project can ask which firms already recognize this signal, what organizational features allow them to do so, and whether those firms capture measurable performance gains from discovering talent that others miss.

5. Recommended framing of the project

Possible research pitch

Employers usually evaluate candidates based on the level of achievement visible on a resume. But the same achievement can require very different amounts of progress depending on a candidate's starting point. We ask whether "distance travelled" - achievement relative to prior opportunity - contains information about latent talent that employers fail to use. We first build an opportunity-adjusted measure from application data and validate it against subsequent worker performance. We then test whether employers underweight this signal and whether a structured second-look tool can identify high-potential candidates who would otherwise be

screened out. Finally, we study firm heterogeneity: which firms are already more likely to hire high-distance-travelled candidates, what features of their screening and organization explain this behavior, and whether these firms obtain better outcomes from doing so.

Connection to the gender-misallocation paper. The existing paper shows that barriers can change the latent composition of the observed talent pool. The new project asks whether employers can detect that selection at the individual level using information embedded in a candidate's path - and why some firms may be substantially better at doing so than others.

References

Agan, Amanda, and Sonja Starr (2018). "Ban the Box, Criminal Records, and Racial Discrimination: A Field Experiment." Quarterly Journal of Economics 133(1): 191–235.

Altonji, Joseph G., and Charles R. Pierret (2001). "Employer Learning and Statistical Discrimination." Quarterly Journal of Economics 116(1): 313–350.

Arcidiacono, Peter, Patrick Bayer, and Aurel Hizmo (2010). "Beyond Signaling and Human Capital: Education and the Revelation of Ability." AEJ: Applied Economics 2(4): 76–104.

Ashraf, Nava, Oriana Bandiera, Virginia Minni, and Víctor Quintas-Martínez (2025). "Gender Gaps across the Spectrum of Development: Local Talent and Firm Productivity." Working paper, R&R at Review of Economics and Statistics.

Autor, David H., and David Scarborough (2008). "Does Job Testing Harm Minority Workers? Evidence from Retail Establishments." Quarterly Journal of Economics 123(1): 219–277.

Bartoš, Vojtěch, Michal Bauer, Julie Chytilová, and Filip Matějka (2016). "Attention Discrimination: Theory and Field Experiments with Monitoring Information Acquisition." American Economic Review 106(6): 1437–1475.

Bastedo, Michael N., et al. (2022). "Admitting Students in Context: Field Experiments on Information Dashboards in College Admissions." Journal of Higher Education 93(3): 327–374.

Behaghel, Luc, Bruno Crépon, and Thomas Le Barbanchon (2015). "Unintended Effects of Anonymous Résumés." American Economic Journal: Applied Economics 7(3): 1–27.

Farber, Henry S., and Robert Gibbons (1996). "Learning and Wage Dynamics." Quarterly Journal of Economics 111(4): 1007–1047.

Hoffman, Mitchell, Lisa B. Kahn, and Danielle Li (2018). "Discretion in Hiring." Quarterly Journal of Economics 133(2): 765–800.

Kessler, Judd B., Corinne Low, and Colin D. Sullivan (2019). "Incentivized Resume Rating: Eliciting Employer Preferences without Deception." American Economic Review 109(11): 3713–3744.

Lange, Fabian (2007). "The Speed of Employer Learning." Journal of Labor Economics 25(1): 1–35.

Li, Danielle, Lindsey Raymond, and Peter Bergman (2026). "Hiring as Exploration." Review of Economic Studies 93(2): 1200–1240.

Mulligan, Casey B., and Yona Rubinstein (2008). "Selection, Investment, and Women's Relative Wages Over Time." Quarterly Journal of Economics 123(3): 1061–1110.

Olivetti, Claudia, and Barbara Petrongolo (2008). "Unequal Pay or Unequal Employment? A Cross-Country Analysis of Gender Gaps." Journal of Labor Economics 26(4): 621–654.

Pallais, Amanda (2014). "Inefficient Hiring in Entry-Level Labor Markets." American Economic Review 104(11): 3565–3599.

Smith, Jeremy, and Robin Naylor (2005). "Schooling Effects on Subsequent University Performance: Evidence for the UK University Population." Economics of Education Review 24(5): 549–562.